

Bioprocess Equations

POV me making my box drawings pink so the censor knows I'm a woman
in STEM



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(Hopefully) All relevant equations for the Bioprocess Technology Exam

1. General mass balance equation

$$\text{Accumulation} = \text{In} - \text{Out} + \text{Generation} - \text{Consumption}$$

Or at steady state

$$\text{In} + \text{Generation} = \text{Out} + \text{Consumption}$$

Where:

- Accumulation = change of material in the system over time (kg s^{-1} , mol s^{-1})
- In = material entering the system (kg s^{-1} , mol s^{-1})
- Out = material leaving the system (kg s^{-1} , mol s^{-1})
- Generation = material formed by reaction (kg s^{-1} , mol s^{-1})
- Consumption = material consumed by reaction (kg s^{-1} , mol s^{-1})

This is used for:

Setting up mass and mole balances in reactors, bioprocesses, and flow systems. It is one of the most important engineering equations because almost all bioprocess calculations are based on balancing what enters, leaves, is produced, and is consumed inside the system.

When to use this:

Use when material enters/leaves a system and something is produced or consumed inside the reactor. Typical trigger words: balance, inlet/outlet, accumulation, steady state, batch, CSTR, overall reactor analysis.

2. Molar conversion equation

$$n = \frac{m}{M_r}$$

Or

$$m = n \cdot M_r$$

Where:

- n = number of moles (mol)
- m = mass (g or kg)
- M_r = molar mass (g mol^{-1})

This is used for:

Converting between mass and moles in stoichiometric and reaction balance problems. It is commonly needed in fermentation and yield calculations where reactions must be balanced on a molar basis.

When to use this:

Use when converting between mass and moles in stoichiometric calculations. Trigger words: molecular weight, mol, stoichiometry, reaction equation, yield in molar units.

3. Mass calculations from concentration.

$$m = CV$$

Where:

- m = total mass (g or kg)
- C = concentration (g L^{-1})
- V = liquid volume (L)

This is used for:

Mass balances in bioprocesses often require conversion from concentration to total mass.

This equation is used constantly in bioprocess calculations because reactor measurements are usually reported as concentrations, while production capacity is evaluated as total mass of biomass or product.

When to use this:

Use when total mass must be determined from concentration and reactor volume. Trigger words: total biomass, product amount, reactor contains, g/L and volume given.

Examples:

$$m_X = X \cdot V$$

$$m_P = P \cdot V$$

Where:

- m_X = biomass mass
- X = biomass concentration
- m_P = product mass
- P = product concentration

4. Atomic balance equations

$$C_{in} = C_{out}$$

$$H_{in} = H_{out}$$

$$O_{in} = O_{out}$$

$$N_{in} = N_{out}$$

Where:

- C, H, O, N = total number of atoms of each element

This is used for:

Balancing stoichiometric equations and solving for unknown stoichiometric coefficients. Commonly used in microbial growth and fermentation reaction problems.

When to use this:

Use when balancing unknown stoichiometric coefficients in microbial reactions. Trigger words: balance the reaction, elemental composition, carbon balance, universal biomass composition, CHON balance.

IMPORTANT:

If "The universal description is assumed for the cell composition" is mentioned – it is " $CH_{1.8}O_{0.5}N_{0.2}$ " – sometimes phosphorus or sulfur may also be included in more detailed formulations, for example: " $CH_{1.8}O_{0.5}N_{0.2}P_{0.02}$ ", " $CH_{1.8}O_{0.5}N_{0.2}S_{0.01}$ " or " $CH_{1.8}O_{0.5}N_{0.2}P_{0.02}S_{0.01}$ ". But if they just give "the universal cell composition and do not mention S or P then use " $CH_{1.8}O_{0.5}N_{0.2}$ " as default.

This is made with reference to carbon, which is what they usually want you to do.

This is used for:

The universal biomass formula is important because it provides a standardised representation of microbial cell composition that can be used when the exact composition is unknown. It allows stoichiometric growth reactions and elemental balances for carbon, hydrogen, oxygen, nitrogen, phosphorus, and sulfur to be calculated in bioprocess engineering problems.

5. Biomass/substrate/product yield equations.

Biomass yield on substrate:

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S}$$

The inverse relationship is:

$$Y_{\frac{S}{X}} = \frac{1}{Y_{\frac{X}{S}}}$$

Product yield on biomass:

$$Y_{\frac{P}{X}} = \frac{\Delta P}{\Delta X}$$

It is used when product formation depends on biomass production.

Product yield on substrate:

$$Y_{\frac{P}{S}} = \frac{\Delta P}{\Delta S}$$

It is used to determine how much product is obtained per gram of substrate.

Where:

- $Y_{X/S}$ = biomass yield coefficient (g biomass g⁻¹ substrate)
- $Y_{S/X}$ = amount of substrate required to produce one gram of biomass (g substrate g⁻¹ biomass)
- $Y_{P/X}$ = product yield on biomass (g product g⁻¹ biomass)
- $Y_{P/S}$ = product yield on substrate (g product g⁻¹ substrate)
- ΔX = biomass formed (g)
- ΔS = substrate consumed (g)
- ΔP = product formed (g)

This is used for:

Yield coefficients are used to describe how efficiently substrate is converted into biomass or product during a bioprocess. They are fundamental in stoichiometric calculations, process optimization, and estimating substrate requirements or product formation.

When to use this:

Use when calculating how efficiently substrate is converted into biomass or product. Trigger words: yield coefficient, g biomass per g glucose, substrate consumption, product formation, conversion efficiency.

6. Cylindrical vessel volume – use when the reactor has a constant diameter.

$$V = \pi \cdot r^2 \cdot h$$

Or

$$V = \frac{\pi \cdot D^2}{4} \cdot h$$

Where:

- V = volume (m³)
- r = radius (m)
- D = diameter (m)
- h = liquid height (m)

This is used for:

Calculating the liquid or working volume of a stirred tank bioreactor. This is an important one as most industrial bioreactors are cylindrical.

When to use this:

Use for standard stirred tank bioreactors with cylindrical geometry. Trigger words: reactor diameter, liquid height, working volume, tank geometry.

7. Spherical vessel volume – sometimes used for storage tanks or bubbles.

$$V = \frac{4}{3} \cdot \pi \cdot r^3$$

Where:

- V = volume (m³)
- r = sphere radius (m)

This is used for:

Calculating the volume of spherical tanks or bubbles. Important because spheres are sometimes used in storage systems and bubble calculations.

When to use this:

Use for spherical tanks, bubbles, or gas pockets. Trigger words: sphere, bubble volume, spherical storage tank.

8. Cone-shaped bottom or top volume – very common in industrial fermenters because many vessels have conical bottoms for drainage.

$$V = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$$

Where:

- V = volume (m³)
- r = radius (m)
- h = cone height (m)
- Sometimes total reactor volume becomes:

$$V_{total} = V_{cylinder} + V_{cone}$$

This is used for:

Calculating cone-shaped reactor bottoms or tops. Important because many fermenters have conical bottoms for drainage and harvesting.

When to use this:

Use when fermenters have conical bottoms or tops. Trigger words: cone bottom, drainage section, conical geometry, total reactor volume.

9. Hemispherical bottom volume – common in real stirred tank reactors.

$$V = \frac{2}{3} \cdot \pi \cdot r^3$$

Where:

- V = volume (m³)
- r = hemisphere radius (m)

This is used for:

Calculating reactor bottom with hemispherical geometry. Important because real industrial tanks often have rounded bottoms for better mixing and cleaning.

When to use this:

Use for rounded-bottom reactor geometries. Trigger words: hemispherical bottom, rounded tank, curved reactor base.

10. Annular volume / jacket calculations volume– for heat transfer jackets or pipe flow.

$$V = \pi \cdot h \cdot (r_o^2 - r_i^2)$$

Where:

- V = annular volume (m³)

- h = height or length (m)
- r_o = outer radius (m)
- r_i = inner radius (m)

This is used for:

Calculating the volume inside the reactor jackets or between pipes. It is used in heat-transfer and flow calculations.

When to use this:

Use for reactor jackets or pipe annulus calculations. Trigger words: cooling jacket, heat transfer, space between pipes, annular region.

11. Working volume vs total volume.

$$V_{working} = f \cdot V_{total}$$

Where:

- $V_{working}$ = working liquid volume (m³)
- V_{total} = total reactor volume (m³)
- f = filling fraction (dimensionless)

This is used for:

When the reactor is not completely filled. Important because bioreactors require headspace for aeration, foam and mixing.

When to use this:

Use when reactors are only partially filled. Trigger words: fill fraction, headspace, working volume, usable reactor volume.

12. Gas hold-up problems volume – sometimes only the liquid phase counts.

$$V_L = V_{total}(1 - \varepsilon_G)$$

Where:

- V_L = liquid volume (m³)
- V_{total} = total reactor volume (m³)
- ε_G = gas hold-up fraction (dimensionless)

This is used for:

Gas bubbles occupy part of the reactor volume. Important because only the liquid phase is available for cells and dissolved compounds.

When to use this:

Use when gas bubbles occupy reactor volume and only liquid volume is biologically active. Trigger words: gas hold-up, aerated reactor, bubble fraction, liquid phase volume.

13. Exponential growth equation for microorganisms

$$X(t) = X(0) \cdot e^{\mu \cdot t}$$

Where:

- $X(t)$ = biomass concentration at time t (g L⁻¹)
- $X(0)$ = initial biomass concentration (g L⁻¹)
- μ = specific growth rate (h⁻¹)
- t = cultivation time (h)

This is used for:

Describing microbial growth during the exponential growth phase in batch cultures. Typical exam questions use this equation to calculate biomass concentration after a certain cultivation time when exponential growth is assumed.

When to use this:

Use during exponential phase batch growth when μ is constant. Trigger words: exponential growth, log phase, batch culture, predict biomass after time t .

14. Specific growth rate isolated from the exponential growth equation

$$\mu = \frac{\ln\left(\frac{X(t)}{X(0)}\right)}{t}$$

Where:

- μ = specific growth rate (h^{-1})
- $X(t)$ = biomass concentration at time t (g L^{-1})
- $X(0)$ = initial biomass concentration (g L^{-1})
- t = cultivation time (h)

This is used for:

Calculating the microbial growth rate from biomass measurements over time. It is commonly used in fermentation kinetics and growth-characterization problems.

When to use this:

Use when biomass concentrations at two time points are known and μ (specific growth rate) must be calculated. Trigger words: determine growth rate, calculate μ , growth kinetics.

15. Time isolated from the exponential growth rate

$$t = \frac{\ln\left(\frac{X(t)}{X(0)}\right)}{\mu}$$

Where:

- t = cultivation time (h)
- $X(t)$ = biomass concentration at time t (g L^{-1})
- $X(0)$ = initial biomass concentration (g L^{-1})
- μ = specific growth rate (h^{-1})

This is used for:

Calculating the time required to reach a desired biomass concentration during exponential growth. Typical questions involve estimating fermentation times in batch cultures.

When to use this:

Use when estimating cultivation time required to reach a target biomass concentration. Trigger words: how long, time required, reach biomass concentration.

16. Doubling time equation

$$t_d = \frac{\ln 2}{\mu}$$

Where:

- t_d = doubling time (h)
- μ = specific growth rate (h^{-1})

This is used for:

Calculating the time required for a microbial population to double during exponential growth. Typical questions involve comparing microbial growth rates.

When to use this:

Use when determining how long cells take to double during exponential growth. Trigger words: doubling time, generation time, cell duplication.

17. Substrate-limited growth equation

$$\mu = \mu_{max} \frac{S}{K_S + S}$$

Where:

- μ = specific growth rate (h^{-1})
- μ_{max} = maximum specific growth rate (h^{-1})
- S = substrate concentration (g L^{-1})
- K_S = Monod constant (g L^{-1})

This is used for:

Describing how microbial growth depends on substrate concentration. Typical exam questions use this equation in chemostat problems and substrate-limited growth kinetics.

When to use this:

Use when microbial growth depends on limiting substrate concentration. Trigger words: Monod kinetics, substrate-limited growth, chemostat kinetics, μ_{max} , K_S .

18. Specific growth rate from biomass balance in chemostat

$$\mu = D$$

Where:

- μ = specific growth rate (h^{-1})
- D = dilution rate (h^{-1})

This is used for:

Describing steady-state operation in continuous cultures. Typical chemostat questions use this equation because the microbial growth rate becomes equal to the dilution rate at steady state.

When to use this:

Use specifically for chemostats operating at steady state. Trigger words: continuous culture, steady-state chemostat, $\mu = D$, continuous reactor.

19. Dilution rate (continuous culture)

$$D = \frac{F}{V}$$

Where:

- D = dilution rate (h^{-1})
- F = volumetric flow rate (L h^{-1}) (can also have the letter Q)
- V = reactor working volume (L).

How to obtain the volumetric flow rate F:

The volumetric flow rate is the volume of liquid entering (or leaving) the reactor per unit time:

$$F = \frac{\Delta V}{\Delta t}$$

Where:

- F = volumetric flow rate (L h^{-1}) (can also have the letter Q)
- ΔV is the volume added (or leaving)

- Δt is the time interval.

This is used for:

This equation is used in chemostat and continuous bioreactor problems because the dilution rate controls nutrient supply, cell removal, and the microbial growth rate at steady state. It is also used to determine whether washout occurs.

Interpretation of dilution rate variables:

- Higher $F \rightarrow$ nutrients enter faster
- Higher $F \rightarrow$ cells are removed faster
- Higher D generally increases growth rate until washout occurs
- Lower D gives longer residence time and slower growth conditions

When to use this:

Use in chemostat and continuous culture calculations. Trigger words: continuous culture, washout, feed rate, dilution rate, steady state.

IMPORTANT

At steady state in a chemostat:

$$\mu = D$$

Where:

- μ = specific growth rate (h^{-1})
- D = dilution rate (h^{-1})

This relationship is important because the microbial growth rate becomes controlled directly by the dilution rate.

If:

$$D > \mu_{max}$$

washout occurs because cells are removed from the reactor faster than they can reproduce.

Where:

- μ_{max} = maximum specific growth rate (h^{-1})

20. Residence time.

$$\tau = \frac{V}{F}$$

Where:

- τ = residence time (h)
- V = reactor volume (L or m^3)
- F = volumetric flow rate (L h^{-1} or $\text{m}^3 \text{h}^{-1}$) (can also have the letter Q)

This is used for:

Residence time describes the average time liquid remains inside a continuous reactor. It is important because it affects substrate conversion, productivity, and steady-state behavior.

A larger residence time means the culture remains longer inside the reactor, generally allowing more complete substrate utilization.

When to use this:

Use for continuous reactors to determine average liquid retention time. Trigger words: residence time, retention time, continuous reactor, average time in reactor.

21. Volumetric flow rate equation

$$Q = A \cdot v$$

And

$$A = \frac{\pi \cdot D^2}{4}$$

Or combined:

$$Q = \frac{\pi \cdot D^2}{4} \cdot v$$

Where:

- Q = volumetric flow rate (m^3s^{-1}) (can also have the letter F)
- A = cross-sectional area (m^2)
- v = fluid velocity (m s^{-1})
- D = pipe diameter (m)

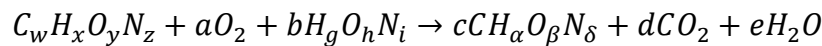
This is used for:

Calculating how much liquid flows through a pipe per unit time. Commonly used in transport, pump, and tank-emptying calculations.

When to use this:

Use when calculating fluid transport through pipes or pumps. Trigger words: flow rate, pipe velocity, cross-sectional area, liquid transfer.

22. General aerobic microbial growth reaction



Where:

- $C_w H_x O_y N_z$ = substrate empirical formula
- $H_g O_h N_i$ = nitrogen source empirical formula
- $C H_\alpha O_\beta N_\delta$ = biomass empirical formula (Cmol)
- a, b, c, d, e = stoichiometric coefficients (mol mol^{-1} substrate), except c , which is (Cmol biomass mol^{-1} substrate)
- $w, x, y, z, g, h, i, \alpha, \beta, \delta$ = empirical formula indices (dimensionless)

This is used for:

Setting up stoichiometric equations for aerobic microbial growth with a carbon source, oxygen, a nitrogen source, biomass, carbon dioxide, and water.

When to use this:

Use when asked to set up or balance an aerobic microbial growth reaction. If the nitrogen source is ammonia, use NH_3 , meaning $g = 3$, $h = 0$, and $i = 1$. Trigger words: aerobic growth, biomass formation, nitrogen source, ammonia, stoichiometric coefficients, oxygen consumption, carbon dioxide formation, elemental balances.

23. Biomass coefficient from biomass yield

For biomass produced per mol substrate:

$$m_X = Y_X \cdot \frac{M_S}{S}$$

For biomass stoichiometric coefficient:

$$c = \frac{m_X}{M_X}$$

Combined equation:

$$c = \frac{Y_X \cdot M_S}{M_X}$$

Where:

- m_X = biomass produced per mol substrate (g biomass mol⁻¹ substrate)
- $Y_{X/S}$ = biomass yield on substrate (g biomass g⁻¹ substrate)
- M_S = molar mass of substrate (g mol⁻¹)
- M_X = molar mass of biomass empirical formula (g Cmol⁻¹)
- c = biomass stoichiometric coefficient (Cmol biomass mol⁻¹ substrate)

This is used for:

Finding the biomass stoichiometric coefficient from a given biomass yield. This is often the first step before solving the remaining stoichiometric coefficients.

When to use this:

Use when a biomass yield is given and the stoichiometric coefficient for biomass is unknown. Trigger words: biomass yield, $Y_{X/S}$, g biomass per g substrate, find biomass coefficient, stoichiometric coefficient for cells.

24. Biomass molar mass from empirical formula

$$M_X = 12 + \alpha + 16 \cdot \beta + 14 \cdot \delta$$

For universal biomass:

$$CH_{1.8}O_{0.5}N_{0.2}$$

$$M_X = 12 + 1.8 + 16(0.5) + 14(0.2) = 24.6 \frac{g}{Cmol}$$

Where:

- M_X = molar mass of biomass empirical formula (g Cmol⁻¹)
- α = H/C ratio in biomass (dimensionless)
- β = O/C ratio in biomass (dimensionless)
- δ = N/C ratio in biomass (dimensionless)

This is used for:

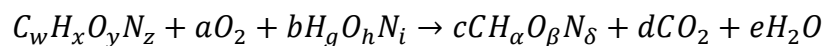
Converting between grams of biomass and Cmol biomass when biomass is written as an empirical formula.

When to use this:

Use when calculating the biomass stoichiometric coefficient from biomass mass or yield. Trigger words: universal biomass formula, $CH_{1.8}O_{0.5}N_{0.2}$, biomass molar mass, Cmol biomass, dry cell weight.

25. Stoichiometric coefficient equations for aerobic growth

For:



Carbon balance:

$$w = c + d$$

Nitrogen balance:

$$z + b \cdot i = c \cdot \delta$$

Hydrogen balance:

$$x + b \cdot g = c \cdot \alpha + 2 \cdot e$$

Oxygen balance:

$$y + 2 \cdot a + b \cdot h = c \cdot \beta + 2 \cdot d + e$$

Useful rearranged forms:

$$d = w - c$$

$$b = \frac{c \cdot \delta - z}{i}$$

$$e = \frac{x + b \cdot g - c \cdot \alpha}{2}$$

$$a = \frac{c \cdot \beta + 2 \cdot d + e - y - b \cdot h}{2}$$

Where:

- w, x, y, z = empirical formula indices for substrate (dimensionless)
- g, h, i = empirical formula indices for nitrogen source (dimensionless)
- α, β, δ = empirical formula indices for biomass (dimensionless)
- a, b, c, d, e = stoichiometric coefficients (mol mol⁻¹ substrate), except c , which is (Cmol biomass mol⁻¹ substrate)

This is used for:

Solving the unknown stoichiometric coefficients in aerobic microbial growth reactions after the biomass coefficient has been found from yield or another given condition.

When to use this:

Use when asked to calculate stoichiometric coefficients for biomass, oxygen, nitrogen source, carbon dioxide, and water. If the nitrogen source is ammonia, use NH_3 , meaning $g = 3, h = 0$, and $i = 1$.

Trigger words: find coefficients, balance aerobic growth reaction, CHON balance, oxygen consumed per mol substrate, CO₂ produced per mol substrate, nitrogen source.

26. Respiratory quotient, RQ

$$RQ = \frac{d}{a}$$

Or generally

$$RQ = \frac{\text{mol CO}_2 \text{ produced}}{\text{mol O}_2 \text{ consumed}}$$

Where:

- RQ = respiratory quotient (dimensionless)
- d = stoichiometric coefficient for carbon dioxide (mol CO₂ mol⁻¹ substrate)
- a = stoichiometric coefficient for oxygen (mol O₂ mol⁻¹ substrate)

This is used for:

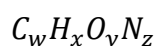
Providing an additional equation when stoichiometric coefficients cannot be solved from elemental balances alone. RQ connects carbon dioxide production to oxygen consumption.

When to use this:

Use when RQ is given or when there are more unknown stoichiometric coefficients than independent elemental balances. Trigger words: respiratory quotient, RQ, CO₂/O₂ ratio, gas analysis, off-gas, oxygen consumption and carbon dioxide production.

27. Degree of reduction for substrate or product

For a compound:



With ammonia as nitrogen reference:

$$\gamma = \frac{4w + x - 2y - 3z}{w}$$

For compounds without nitrogen:

$$\gamma = \frac{4w + x - 2y}{w}$$

Where:

- γ = degree of reduction ($\text{mol e}^- \text{ Cmol}^{-1}$)
- $C_w H_x O_y N_z$ = empirical formula of substrate, biomass or product
- w, x, y, z = empirical formula indices (dimensionless)

This is used for:

Calculating the available electrons per Cmol of a compound. It is used in electron balances, redox balances, theoretical oxygen demand, and maximum theoretical yield calculations.

When to use this:

Use when the question mentions electron balance, degree of reduction, available electrons, reducing power, redox balance, theoretical oxygen demand, or maximum theoretical yield. The formula with $-3z$ is used when nitrogen is referenced to ammonia, which is the common convention in bioprocess stoichiometry. If another nitrogen reference state is specified, follow the formula or convention given in the problem. Trigger words: degree of reduction, electron balance, available electrons, reducing power, redox balance, theoretical oxygen demand, maximum theoretical yield.

28. Electron balance for microbial stoichiometry

For substrate converted into biomass and product

$$w \cdot \gamma_S = c \cdot \gamma_X + p \cdot w_P \cdot \gamma_P + 4 \cdot a$$

If no product is formed:

$$w \cdot \gamma_S = c \cdot \gamma_X + 4 \cdot a$$

Where:

- w = number of carbon atoms in substrate (dimensionless)
- γ_S = degree of reduction of substrate ($\text{mol e}^- \text{ Cmol}^{-1}$)
- c = stoichiometric coefficient for biomass ($\text{Cmol biomass mol}^{-1}$ substrate)
- γ_X = degree of reduction of biomass ($\text{mol e}^- \text{ Cmol}^{-1}$)
- p = stoichiometric coefficient for product ($\text{mol product mol}^{-1}$ substrate)
- w_P = number of carbon atoms in product (dimensionless)
- γ_P = degree of reduction of product ($\text{mol e}^- \text{ Cmol}^{-1}$)
- a = stoichiometric coefficient for oxygen ($\text{mol O}_2 \text{ mol}^{-1}$ substrate)

This is used for:

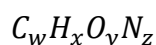
Checking redox consistency and solving microbial stoichiometry when oxygen consumption, biomass formation, or product formation is unknown.

When to use this:

Use when elemental balances alone are not enough or when redox/electron balance is specifically mentioned. Trigger words: electron balance, degree of reduction, available electrons, oxygen demand, redox balance, maximum theoretical yield.

29. Theoretical oxygen demand, ThOD

For complete oxidation of:



To CO_2 , H_2O , and NH_3 :

$$a = w + \frac{x}{4} - \frac{y}{2} - \frac{3 \cdot z}{4}$$

Mass-based theoretical oxygen demand:

$$\text{ThOD} = \frac{a \cdot 32}{M_S}$$

Where:

- a = mol O_2 required per mol substrate ($\text{mol } O_2 \text{ mol}^{-1} \text{ substrate}$)
- w = number of carbon atoms in substrate (dimensionless)
- x = number of hydrogen atoms in substrate (dimensionless)
- y = number of oxygen atoms in substrate (dimensionless)
- z = number of nitrogen atoms in substrate (dimensionless)
- $ThOD$ = theoretical oxygen demand ($\text{g } O_2 \text{ g}^{-1} \text{ substrate}$)
- 32 = molar mass of oxygen (g mol^{-1})
- M_s = molar mass of substrate (g mol^{-1})

This is used for:

Calculating the oxygen required for complete oxidation of a substrate. In bioprocesses, it gives the theoretical maximum oxygen demand if the substrate is fully oxidised.

When to use this:

Use when asked for theoretical oxygen demand or complete oxidation oxygen requirement. Trigger words: ThOD, theoretical oxygen demand, complete oxidation, oxygen required, aerobic oxidation, substrate to CO_2 and H_2O .

30. Oxygen transfer problems – the most important equation in aerobic bioprocesses.

$$OTR = k_L a (C^* - C_L)$$

Where:

- OTR = oxygen transfer rate ($\text{mol L}^{-1} \text{h}^{-1}$)
- $k_L a$ = volumetric oxygen transfer coefficient (h^{-1})
- C^* = saturated dissolved oxygen concentration (mol L^{-1})
- C_L = actual dissolved oxygen concentration (mol L^{-1})

This is used for:

Which is used when calculating oxygen transfer capacity in aerobic bioreactors. It determines how much oxygen can enter the culture broth.

When to use this:

Use when calculating oxygen supply from gas phase into liquid culture. Trigger words: oxygen transfer, $k_L a$, aeration, oxygen supply, OTR.

31. Oxygen uptake rate from specific oxygen demand

$$OUR = q_{O_2} \cdot X$$

Where:

- OUR = oxygen uptake rate ($\text{mol } O_2 \text{ L}^{-1} \text{h}^{-1}$)
- q_{O_2} = specific oxygen uptake rate/demand ($\text{mol } O_2 \text{ g}_X^{-1} \text{h}^{-1}$)
- X = biomass concentration ($\text{g}_X \text{ L}^{-1}$)

This is used for:

Calculating the total oxygen consumption rate per reactor volume from biomass concentration.

When to use this:

Use when the problem gives specific oxygen demand and biomass concentration, or when biomass must first be calculated from substrate yield. Trigger words: specific oxygen demand, specific oxygen uptake rate, q_{O_2} , per gram dry weight per hour.

32. The oxygen uptake rate (OUR) – Oxygen transfer rate (OTR) under oxygen limited conditions

$$OTR = OUR = q_{O_2} \cdot X$$

Where:

- q_{O_2} = specific oxygen uptake rate ($\text{mol g}^{-1}\text{h}^{-1}$)
- X = biomass concentration (g L^{-1})
- OUR = oxygen uptake rate ($\text{mol L}^{-1}\text{h}^{-1}$)
- OTR = oxygen transfer rate ($\text{mol L}^{-1}\text{h}^{-1}$)

This is used for:

This equation is used to calculate the oxygen demand of the culture based on the biomass concentration. It is important because it helps determine whether the reactor can supply enough oxygen to support cell growth.

When to use this:

Use under oxygen-limited or steady-state oxygen conditions where oxygen consumption equals oxygen transfer. Trigger words: oxygen demand, OUR, oxygen-limited growth, oxygen balance.

33. The steady state oxygen balance equation

$$q_{O_2} \cdot X = OTR \rightarrow q_{O_2} \cdot X = K_L a (C^* - C_L)$$

Where:

- OTR = oxygen transfer rate ($\text{mol L}^{-1}\text{h}^{-1}$)
- $k_L a$ = volumetric oxygen transfer coefficient (h^{-1})
- C^* = saturated dissolved oxygen concentration (mol L^{-1})
- C_L = actual dissolved oxygen concentration (mol L^{-1})
- q_{O_2} = specific oxygen uptake rate ($\text{mol g}^{-1}\text{h}^{-1}$)
- X = biomass concentration (g L^{-1})

This is used for:

Which is used when checking if growth is oxygen limited. It balances oxygen demand by cells with oxygen supply by the reactor.

When to use this:

Use to compare microbial oxygen demand with reactor oxygen supply. Trigger words: oxygen limitation check, oxygen balance, can reactor support growth, steady-state oxygen transfer.

34. Critical oxygen transfer coefficient

$$k_L a_{crit} = \frac{q_{O_2} \cdot X}{C^* - C_{crit}}$$

Where:

- $k_L a_{crit}$ = critical volumetric oxygen transfer coefficient (h^{-1})
- q_{O_2} = specific oxygen uptake rate ($\text{mol g}^{-1}\text{h}^{-1}$)
- X = biomass concentration (g L^{-1})
- C^* = saturation dissolved oxygen concentration (mol L^{-1})
- C_{crit} = critical dissolved oxygen concentration (mol L^{-1})

This is used for:

Calculating the minimum oxygen transfer capacity required to prevent oxygen limitation in aerobic cultures. Typical exam questions use this equation to determine whether the reactor can maintain dissolved oxygen above the critical oxygen concentration needed for microbial growth.

When to use this:

Use to determine the minimum oxygen transfer capability needed to avoid oxygen limitation. Trigger words: oxygen limitation, minimum kLa, critical dissolved oxygen, aerobic culture requirement.

35. Maximum biomass concentration

$$X_{max} = \frac{k_L a (C^* - C_L)}{q_{O_2}}$$

Where:

- $k_L a$ = volumetric oxygen transfer coefficient (h^{-1})
- C^* = saturated dissolved oxygen concentration (mol L^{-1})
- C_L = actual dissolved oxygen concentration (mol L^{-1})
- q_{O_2} = specific oxygen uptake rate ($\text{mol g}^{-1} \text{h}^{-1}$)
- X_{max} = maximum biomass concentration (g L^{-1})

This is used for:

This equation is used to calculate the maximum biomass concentration that can be supported by the reactor's oxygen transfer capacity. It is important because oxygen transfer is often the limiting factor in aerobic bioprocesses.

When to use this:

Use when oxygen transfer limits maximum achievable biomass concentration. Trigger words: oxygen-limited biomass, maximum cell density, OTR limitation.

36. Total oxygen consumption rate in reactor

$$\dot{n}_{O_2} = OUR \cdot V$$

Where:

- $\dot{n}_{O_2}$ = total oxygen consumption rate ($\text{mol O}_2 \text{ h}^{-1}$)
- OUR = oxygen uptake rate per volume ($\text{mol O}_2 \text{ L}^{-1} \text{ h}^{-1}$)
- V = reactor/culture volume (L)

This is used for:

Converting oxygen uptake per liter into total oxygen consumption for the full fermenter.

When to use this:

Use when reactor volume is given and you need total oxygen demand or total heat generation. Trigger words: 8 m³ fermenter, total oxygen consumed, whole reactor, total heat released.

37. Biomass concentration from substrate yield

$$X = Y_{X/S} \cdot S$$

Where:

- X = biomass concentration formed ($\text{g}_X \text{ L}^{-1}$)
- $Y_{X/S}$ = biomass yield on substrate ($\text{g}_X \text{ g}_S^{-1}$)
- S = substrate concentration consumed ($\text{g}_S \text{ L}^{-1}$)

This is used for:

Estimating biomass formed if a known amount of substrate is consumed.

When to use this:

Use when the problem says something like 1 g cells produced per 2.2 g substrate consumed or gives a biomass yield. Trigger words: for every g of substrate consumed, complete substrate consumption, biomass yield, dry weight cells.

38. Heat generated from oxygen consumption

$$\dot{Q}_{gen} = OUR \cdot V \cdot \Delta H_{O_2}$$

Where:

- $\dot{Q}_{gen}$ = metabolic heat generation rate (J h^{-1}) or (W)
- OUR = oxygen uptake rate ($\text{mol O}_2 \text{ L}^{-1} \text{ h}^{-1}$)
- V = culture volume (L)
- ΔH_{O_2} = heat released per mole oxygen consumed ($\text{J mol}^{-1} \text{ O}_2$)

This is used for:

Estimating heat production from microbial metabolism based on oxygen consumption.

When to use this:

Use when the problem states heat released per mole of oxygen consumed, e.g. 460 kJ per gmol O_2 .

Trigger words: heat released, oxygen consumed, metabolic heat, temperature must be held constant, cooling requirement.

NOTE:

If your $\dot{Q}_{gen}$ has the unit (J h^{-1}), then divide by 3600 to get W.

39. Unit conversion for metabolic heat

$$\begin{aligned}
 1W &= 1 \frac{\text{J}}{\text{s}} \\
 1h &= 3600 \text{ s} \\
 1 \text{ m}^3 &= 1000L \\
 1 \text{ mmol} &= 10^{-3} \text{ mol} \\
 460 \frac{\text{kJ}}{\text{mol}} &= 460000 \frac{\text{J}}{\text{mol}}
 \end{aligned}$$

This is used for:

Making the units consistent between heat generation and heat removal.

When to use this:

Use almost always in heat-transfer/bioreactor temperature-control problems. Trigger words: kJ, W, h, s, m^3 , L, mmol.

40. Heat balance for cooling sufficiency

$$\dot{Q}_{rem} \geq \dot{Q}_{gen}$$

Where:

- $\dot{Q}_{rem}$ = heat removal capacity (W)
- $\dot{Q}_{gen}$ = metabolic heat generation rate (W)

This is used for:

Checking whether the cooling system can maintain constant fermentation temperature.

If the heat removal capacity is equal to or greater than the metabolic heat generation rate, the cooling system can maintain a constant fermentation temperature. If not, heat accumulates and the fermentation temperature will rise.

When to use this:

Use when the problem asks whether the heat transfer system supports complete substrate consumption or whether temperature can be maintained. Trigger words: does the heat transfer system support, temperature must be held constant, cooling sufficient, complete substrate consumption.

41. Heat transfer rate through cooling surface

$$\dot{Q} = U \cdot A \cdot \Delta T$$

Where:

- $\dot{Q}$ = heat transfer rate ($W = J s^{-1}$)
- U = overall heat transfer coefficient ($W m^{-2} K^{-1}$)
- A = heat transfer area (m^2)
- ΔT = temperature difference between broth and coolant (K or $^{\circ}C$)

This is used for:

Calculating how much heat can be removed from or added to a bioreactor through a jacket, coil, or heat exchanger.

When to use this:

Use when the problem gives a heat transfer coefficient, cooling coil/jacket area, broth temperature, coolant temperature, or asks whether cooling is sufficient. Trigger words: heat transfer coefficient, cooling coil, jacket, temperature control, heat removal, fermentation temperature.

42. Simple temperature driving force

$$\Delta T = T_{broth} - T_{coolant}$$

Where:

- ΔT = temperature difference driving heat transfer (K or $^{\circ}C$)
- T_{broth} = fermentation/broth temperature ($^{\circ}C$)
- $T_{coolant}$ = coolant temperature ($^{\circ}C$)

This is used for:

Estimating the driving force for heat removal when only one coolant temperature is given.

When to use this:

Use when the problem gives the fermentation temperature and only the coolant inlet temperature, without outlet temperature. Trigger words: coolant inlet temperature, held constant at, fermentation temperature, temperature difference.

43. Log mean temperature difference, LMTD

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln \left(\frac{\Delta T_1}{\Delta T_2} \right)}$$

Where:

- ΔT_{lm} = log mean temperature difference (K or $^{\circ}C$)
- ΔT_1 = temperature difference at one end of heat exchanger (K or $^{\circ}C$)
- ΔT_2 = temperature difference at the other end (K or $^{\circ}C$)

This is used for:

Calculating the effective temperature driving force when coolant inlet and outlet temperatures are both known.

When to use this:

Use when both inlet and outlet coolant temperatures are given. Trigger words: coolant outlet temperature, heat exchanger, temperature profile, log mean temperature difference, LMTD.

44. Cooling coil surface area

$$A = \pi \cdot D \cdot L$$

Where:

- A = external heat transfer area of the coil (m^2)
- D = coil pipe diameter (m)
- L = total coil length (m)

This is used for:

Calculating the heat transfer area of a cylindrical cooling coil inside a fermenter.

When to use this:

Use when the problem gives coil length and pipe diameter. Trigger words: helical cooling coil, pipe diameter, coil length, cooling surface area.

45. Jacket heat transfer area for cylindrical vessel

$$A = \pi \cdot D \cdot H$$

Where:

- A = jacket heat transfer area (m^2)
- D = vessel diameter (m)
- H = liquid height in contact with jacket (m)

This is used for:

Estimating heat transfer area for a jacketed cylindrical bioreactor.

When to use this:

Use when the reactor has a cooling/heating jacket rather than an internal coil. Trigger words: cooling jacket, jacketed fermenter, vessel wall heat transfer, liquid height.

46. Required heat transfer area

$$A = \frac{\dot{Q}}{U \cdot \Delta T}$$

Or

$$A = \frac{\dot{Q}}{U \cdot \Delta T_{lm}}$$

Where:

- A = required heat transfer area (m^2)
- $\dot{Q}$ = heat that must be removed or supplied (W)
- U = overall heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
- ΔT = simple temperature difference (K)
- ΔT_{lm} = log mean temperature difference (K)

This is used for:

Sizing a coil, jacket, or heat exchanger.

When to use this:

Use when asked how large the cooling/heating surface must be. Trigger words: required coil length, required heat transfer area, design cooling system, is the coil large enough.

47. Required coil length

$$L = \frac{A}{\pi \cdot D}$$

Where:

- L = required coil length (m)
- A = required heat transfer area (m^2)
- D = coil pipe diameter (m)

This is used for:

Calculating the cooling coil length needed to provide sufficient heat transfer area.

When to use this:

Use when asked for coil length or whether a given coil is sufficient. Trigger words: required coil length, cooling coil diameter, coil sizing.

48. Sensible heat requirement

$$Q = m \cdot c_p \cdot \Delta T$$

Where:

- Q = heat required or removed (J)
- m = mass of liquid/medium (kg)
- c_p = specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
- ΔT = temperature change (K)

This is used for:

Calculating heat needed to warm up or cool down a medium.

When to use this:

Use when a liquid or medium changes temperature. Trigger words: heat from 20°C to 32°C, cool down, warm up, temperature change, specific heat capacity.

49. Heating/cooling rate from utility flow

$$\dot{Q} = \dot{m} \cdot c_p \cdot \Delta T$$

Where:

- $\dot{Q}$ = heat transfer rate ($\text{W} = \text{J s}^{-1}$)
- $\dot{m}$ = mass flow rate of utility fluid (kg s^{-1})
- c_p = specific heat capacity of utility fluid ($\text{J kg}^{-1} \text{K}^{-1}$)
- ΔT = temperature change of utility fluid (K)

This is used for:

Calculating heat removed or supplied by cooling water, steam, or another utility stream.

When to use this:

Use when coolant/heating fluid flowrate and inlet/outlet temperatures are given. Trigger words: cooling water flowrate, water enters at, water leaves at, utility stream, heat removed by water.

50. Coolant flowrate required

$$\dot{m}_{coolant} = \frac{\dot{Q}}{c_p(T_{out} - T_{in})}$$

Where:

- $\dot{m}_{coolant}$ = required coolant mass flow rate (kg s^{-1})
- $\dot{Q}$ = heat that must be removed (W)
- c_p = specific heat capacity of coolant ($\text{J kg}^{-1} \text{K}^{-1}$)
- T_{out} = coolant outlet temperature (K)
- T_{in} = coolant inlet temperature (K)
- $T_{out} - T_{in}$ = coolant temperature increase (K)

This is used for:

Calculating how much cooling water is needed to remove process or metabolic heat.

When to use this:

Use when asked for required cooling water flowrate. Trigger words: cooling water required, water inlet/outlet temperature, remove heat, coolant flow.

51. Reynolds number (Pipe flow)

$$Re = \frac{\rho \cdot u \cdot D}{\mu}$$

Where:

- Re = Reynolds number (dimensionless)
- ρ = fluid density (kg m^{-3})
- u = fluid velocity (m s^{-1})
- D = pipe diameter (m)
- μ = dynamic viscosity (Pa s)

This is used for:

Determining whether flow in a pipe is laminar or turbulent. This is important because flow regime affects friction losses, pressure drop, and mixing behaviour.

When to use this:

Use to determine laminar or turbulent flow in pipes. Trigger words: flow regime, pipe turbulence, pressure loss, laminar flow, turbulent flow.

IMPORTANT:

Reynolds number determination of flow for pipe flow: laminar, transitional or turbulent.

Flow regime:

Reynolds number	Flow regime
$Re < 2300$	Laminar
$2300 < Re < 4000$	Transitional
$Re > 4000$	Turbulent

52. Newton's law of viscosity

$$\tau = \mu \cdot \frac{dv}{dy}$$

Where:

- τ = shear stress (Pa)
- μ = dynamic viscosity (Pa · s)
- $\frac{dv}{dy}$ = velocity gradient / shear rate (s^{-1})

This is used for:

Describing how a Newtonian fluid resists deformation when layers of fluid move at different velocities. It relates shear stress to viscosity and shear rate.

When to use this:

Use when the problem involves viscosity, shear stress, shear rate, or Newtonian fluid behaviour. Trigger words: viscosity, shear stress, shear rate, velocity gradient, Newtonian fluid.

53. Laminar flow in circular tubes / Poiseuille's law

For laminar velocity profile:

$$v(r) = v_{max} \left(1 - \frac{r^2}{R_2^2} \right)$$

For mean velocity:

$$\bar{v} = \frac{Q}{A}$$

And for laminar tube flow:

$$\bar{v} = \frac{1}{2} \cdot v_{max}$$

Or:

$$v_{max} = 2 \cdot \bar{v}$$

For Poiseuille's law:

$$Q = \frac{\Delta P \cdot \pi \cdot R^4}{8 \cdot \mu \cdot L}$$

Rearranged for pressure drop:

$$\Delta P = \frac{8 \cdot \mu \cdot L \cdot Q}{\pi \cdot R^4}$$

For hydraulic resistance:

$$Q = \frac{\Delta P}{R_h}$$
$$R_h = \frac{8 \cdot \mu \cdot L}{\pi \cdot R^4}$$

Using diameter instead of radius:

$$Q = \frac{\pi \cdot D^4 \cdot \Delta P}{128 \cdot \mu \cdot L}$$

Where:

- $v(r)$ = local velocity at radial position r ($m \cdot s^{-1}$)
- v_{max} = maximum velocity at the centre of the tube ($m \cdot s^{-1}$)
- $\bar{v}$ = mean / average velocity ($m \cdot s^{-1}$)
- Q = volumetric flowrate ($m^3 \cdot s^{-1}$)
- A = cross-sectional area (m^2)
- ΔP = pressure difference / pressure drop (Pa)
- R = tube radius (m)
- D = tube diameter (m)
- μ = dynamic viscosity ($Pa \cdot s$)
- L = tube length (m)
- R_h = hydraulic resistance to flow ($Pa \cdot s \cdot m^{-3}$)

This is used for:

Calculating velocity profile, mean velocity, flowrate, pressure drop, or hydraulic resistance for fully developed laminar flow in a circular tube.

When to use this:

Use when the flow is laminar in a circular pipe/tube and the problem involves pressure-driven flow.

Trigger words: Poiseuille, laminar tube flow, capillary flow, pressure drop, pressure-driven flow, hydraulic resistance, parabolic velocity profile, radius to the fourth power.

IMPORTANT

Poiseuille's law is only valid for laminar flow in a circular tube. If the flow regime is not stated, check Reynolds number first.

54. Bernoulli equation

$$\frac{P_1}{\rho \cdot g} + z_1 + \frac{v_1^2}{2 \cdot g} = \frac{P_2}{\rho \cdot g} + z_2 + \frac{v_2^2}{2 \cdot g} + h_L$$

Where:

- P = pressure (Pa)
- ρ = fluid density (kg m^{-3})
- g = gravitational acceleration (m s^{-2})
- z = elevation (m)
- v = fluid velocity (m s^{-1})
- h_L = head loss (m)

This is used for:

Solving fluid flow and pump problems by relating pressure, elevation, velocity, and friction losses. Typical exam problems involve pipes, tanks, and flowing fluids.

When to use this:

Use when relating pressure, velocity, and height in flowing fluids. Trigger words: fluid energy balance, pressure drop, pipe flow, pump system.

55. Torricelli's law

$$v = \sqrt{2 \cdot g \cdot h}$$

Where:

- v = outlet velocity (m s^{-1})
- g = gravitational acceleration (m s^{-2})
- h = height difference (m)

This is used for:

Calculating outlet velocity from tanks under gravity when friction is neglected. It is a simplified form of Bernoulli's equation commonly used in draining problems.

When to use this:

Use for gravity-driven tank draining when friction losses are neglected. Trigger words: draining tank, outlet velocity, gravity flow, ideal discharge.

56. Venturimeter flow equation

$$v_2 = \sqrt{\frac{2 \cdot \Delta P}{\rho \cdot \left(1 - \left(\frac{A_2}{A_1}\right)^2\right)}}$$

And

$$Q = A_2 \cdot v_2$$

Combined:

$$Q = A_2 \cdot \sqrt{\frac{2 \cdot \Delta P}{\rho \cdot \left(1 - \left(\frac{A_2}{A_1}\right)^2\right)}}$$

Where:

- v_2 = velocity at the narrow section / throat ($\text{m} \cdot \text{s}^{-1}$)
- Q = volumetric flowrate ($\text{m}^3 \cdot \text{s}^{-1}$)
- A_1 = cross-sectional area of the wide section (m^2)
- A_2 = cross-sectional area of the narrow section / throat (m^2)
- $\Delta P = P_1 - P_2$ = pressure difference between wide and narrow section (Pa)
- ρ = fluid density ($\text{kg} \cdot \text{m}^{-3}$)

This is used for:

Calculating fluid velocity or volumetric flowrate from a measured pressure difference across a constriction in a pipe.

When to use this:

Use when the problem involves a Venturimeter or a constricted pipe used to measure flowrate from pressure difference. Trigger words: Venturi, Venturimeter, throat, constriction, pressure difference, flow measurement, Bernoulli, continuity.

57. Relative roughness

$$\frac{\varepsilon}{D}$$

Where:

- ε = absolute pipe roughness (m)
- D = pipe diameter (m)
- $\frac{\varepsilon}{D}$ = relative roughness (*dimensionless*)

This is used for:

Describing how rough the inside of a pipe is compared with the pipe diameter. Relative roughness is used together with Reynolds number to find the friction factor for pipe flow.

When to use this:

Use when the problem gives pipe roughness, absolute roughness, pipe material, wall roughness, or asks for friction factor using a Moody diagram. Trigger words: roughness, relative roughness, absolute roughness, pipe material, pipe wall, Moody diagram, friction factor, turbulent pipe flow.

IMPORTANT:

Relative roughness is dimensionless because both ε and D have units of length.

58. Moody diagram

Reynolds number:

$$Re = \frac{\rho \cdot v \cdot D}{\mu}$$

Relative roughness:

$$\frac{\varepsilon}{D}$$

Darcy friction factor:

$$f_D = f_D \left(Re, \frac{\varepsilon}{D} \right)$$

Where:

- Re = Reynolds number for pipe flow (*dimensionless*)
- ρ = fluid density ($kg \cdot m^{-3}$)
- v = average fluid velocity in the pipe ($m \cdot s^{-1}$)
- D = pipe diameter (m)
- μ = dynamic viscosity ($Pa \cdot s$)
- ε = absolute pipe roughness (m)
- $\frac{\varepsilon}{D}$ = relative roughness (*dimensionless*)
- f_D = Darcy friction factor read from the Moody diagram (*dimensionless*)

This is used for:

Finding the Darcy friction factor for pipe flow from the Reynolds number and the relative roughness. The friction factor is then used in Darcy–Weisbach to calculate head loss or pressure drop.

When to use this:

Use when the problem gives pipe diameter, pipe roughness/material, Reynolds number, or asks for friction factor using a Moody chart/diagram. Trigger words: Moody diagram, Moody chart, friction factor, relative roughness, pipe roughness, turbulent pipe flow, head loss, pressure drop, Darcy-Weisbach.

How to use:

- 1) Calculate Reynolds number:

$$Re = \frac{\rho \cdot v \cdot D}{\mu}$$

- 2) Calculate relative roughness:

$$\frac{\varepsilon}{D}$$

- 3) Find Re on the x-axis of the Moody diagram
- 4) Follow upward until you reach the curve for the correct $\frac{\varepsilon}{D}$
- 5) Read the Darcy friction factor f_D from the y-axis.
- 6) Use f_D in Darcy-Weisbach:

$$h_f = f_D \cdot \frac{L}{D} \cdot \frac{v^2}{2 \cdot g}$$

Or:

$$\Delta P = f_D \cdot \frac{L}{D} \cdot \frac{\rho \cdot v^2}{2}$$

IMPORTANT:

The Moody diagram normally gives the Darcy friction factor f_D , not the Fanning friction factor. If the problem gives the Fanning friction factor f_F , convert using:

$$f_D = 4 \cdot f_F$$

Or

$$f_F = \frac{f_D}{4}$$

For laminar flow:

$$f_D = \frac{64}{Re}$$

For transitional pipe flow:

$$2100 < Re < 4000$$

the friction factor is uncertain, so avoid using a precise value unless the problem gives a method, Moody chart value, or asks for an estimate.

59. Fanning friction factor

For laminar pipe flow

$$f_F = \frac{16}{Re}$$

Relationship between Darcy friction factor and fanning friction factor:

$$f_D = 4 \cdot f_F$$

Or

$$f_F = \frac{f_D}{4}$$

Where:

- f_F = Fanning friction factor (*dimensionless*)
- f_D = Darcy friction factor (*dimensionless*)

- Re = Reynolds number (*dimensionless*)

This is used for:

Calculating or converting friction factors in pipe-flow pressure-drop and head-loss problems. Some equations and charts use the Darcy friction factor, while others use the Fanning friction factor.

When to use this:

Use when the problem specifically says Fanning friction factor, skin friction coefficient, or gives a friction factor that must be converted before using Darcy–Weisbach. Trigger words: Fanning friction factor, Darcy friction factor, friction factor conversion, $f_D = 4f_F$, $16/Re$, $64/Re$, skin friction coefficient.

IMPORTANT:

Do not mix up Darcy and Fanning friction factors.

For laminar flow:

$$f_F = \frac{16}{Re}$$

But:

$$f_D = \frac{64}{Re}$$

Darcy–Weisbach normally uses the Darcy friction factor f_D , unless the problem explicitly states otherwise.

60. Fanning pressure-drop equation

$$\Delta P = 4 \cdot f_F \cdot \frac{L}{D} \cdot \frac{\rho \cdot v^2}{2}$$

Equivalent form:

$$\Delta P = 2 \cdot f_F \cdot \frac{L}{D} \cdot \rho \cdot v^2$$

For head loss:

$$h_f = 4 \cdot f_F \cdot \frac{L}{D} \cdot \frac{v^2}{2 \cdot g}$$

Where:

- ΔP = pressure drop due to pipe friction (Pa)
- h_f = friction head loss (m)
- f_F = Fanning friction factor (*dimensionless*)
- L = pipe length (m)
- D = pipe diameter (m)
- ρ = fluid density ($kg \cdot m^{-3}$)
- v = average fluid velocity ($m \cdot s^{-1}$)
- g = gravitational acceleration ($m \cdot s^{-2}$)

This is used for:

Calculating pressure drop or head loss in pipe flow when the Fanning friction factor is used instead of the Darcy friction factor.

When to use this:

Use when the problem specifically gives or asks for the Fanning friction factor and asks for pressure drop, head loss, or friction loss. Trigger words: Fanning equation, Fanning pressure drop, Fanning friction factor, pressure drop, pipe friction, friction loss, $4f$, $2f\rho v^2$.

IMPORTANT:

This equation uses the Fanning friction factor f_F .

Do not use the Darcy friction factor directly in this equation unless you convert it first.

The relationship is:

$$f_D = 4f_F$$

So the Darcy pressure-drop equation:

$$\Delta P = f_D \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$$

is equivalent to the Fanning pressure-drop equation because:

$$f_D = 4f_F$$

61. Darcy-Weisbach equation

For head loss:

$$h_f = f \cdot \frac{L}{D} \cdot \frac{v^2}{2 \cdot g}$$

For pressure drop:

$$\Delta P = f_D \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$$

Where:

- h_f = friction head loss (m)
- ΔP = pressure drop due to pipe friction (Pa)
- f_D = Darcy friction factor (dimensionless)
- L = pipe length (m)
- D = pipe diameter (m)
- v = average fluid velocity ($m \cdot s^{-1}$)
- g = gravitational acceleration ($m \cdot s^{-2}$)
- ρ = fluid density ($kg \cdot m^{-3}$)

This is used for:

Calculating energy and pressure losses caused by pipe friction. Typical questions involve pump requirements and flow losses in piping systems. And calculating head loss or pressure drop due to friction in pipe flow.

When to use this:

Use when calculating pressure or energy losses due to pipe friction. Trigger words: friction loss, pressure drop, pipe resistance, head loss.

Use when calculating pressure or energy losses due to pipe friction. Trigger words: Darcy–Weisbach, Darcy friction factor, pipe friction, head loss, pressure drop, pump requirement, pipe flow, friction loss.

IMPORTANT:

This equation uses the Darcy friction factor f_D .

If the problem gives the Fanning friction factor f_F , convert first:

$$f_D = 4f_F$$

Do not use the Fanning friction factor directly in the Darcy–Weisbach equation unless the equation has specifically been written in Fanning form.

62. Minor head loss from fittings

$$h_m = K \cdot \frac{v^2}{2 \cdot g}$$

Where:

- h_m = minor head loss from fitting (m)
- K = loss coefficient for fitting (dimensionless)
- v = fluid velocity (m s^{-1})
- g = gravitational acceleration (m s^{-2})

This is used for:

Calculating extra head loss caused by fittings, valves, bends, elbows, entrances, exits, contractions, and expansions in a pipe system.

When to use this:

Use when the problem includes pipe fittings or local flow disturbances in addition to straight pipe friction. Trigger words: valve, bend, elbow, entrance, exit, contraction, expansion, fitting loss, minor loss, local loss.

63. Total head loss in pipe systems

$$h_L = h_f + \sum h_m$$

Or:

$$h_L = f \cdot \frac{L}{D} \cdot \frac{v^2}{2 \cdot g} + \sum K \cdot \frac{v^2}{2 \cdot g}$$

Where:

- h_L = total head loss (m)
- h_f = major head loss from pipe friction (m)
- h_m = minor head loss from fittings (m)
- f = Darcy friction factor (dimensionless)
- L = pipe length (m)
- D = pipe diameter (m)
- K = fitting loss coefficient (dimensionless)
- v = fluid velocity (m s^{-1})
- g = gravitational acceleration (m s^{-2})

This is used for:

Calculating the total energy loss in a pipe system due to both straight pipe friction and fittings.

When to use this:

Use when a pipe system includes both pipe length and fittings, and the total pressure drop or pump requirement must be calculated. Trigger words: total head loss, pipe friction, fittings, valves, bends, pressure drop, pump requirement, piping system.

64. Pressure drop from head loss

$$\Delta P = \rho \cdot g \cdot h_L$$

Where:

- ΔP = pressure drop (Pa)
- ρ = fluid density (kg m^{-3})
- g = gravitational acceleration (m s^{-2})
- h_L = head loss (m)

This is used for:

Converting friction/head loss expressed as meters of liquid into pressure drop.

When to use this:

Use when a problem gives head loss and asks for pressure drop, pump pressure, or pressure requirement. Trigger words: head loss, pressure drop, m water column, pump pressure.

65. Mechanical energy balance for pumps / Specific shaft work from the mechanical energy balance

$$w_s = \frac{\Delta P}{\rho} + \frac{v_2^2 - v_1^2}{2} + g \cdot (z_2 - z_1) + e_f$$

Where:

- w_s = specific shaft work (J kg^{-1})
- ΔP = pressure difference (Pa)
- ρ = fluid density (kg m^{-3})
- v = fluid velocity (m s^{-1})
- g = gravitational acceleration (m s^{-2})
- z = elevation (m)
- e_f = frictional energy loss (J kg^{-1})

This is used for:

Calculating the work required to move fluids through systems with elevation changes, velocity changes, pressure differences, and friction losses. Commonly used in pump and piping problems.

When to use this:

Use when calculating work required to move fluid through a system with pressure, elevation, or velocity changes. Trigger words: pump work, mechanical energy balance, shaft work, fluid transport energy.

IMPORTANT:

This equation is used to calculate the pump work required per kg of fluid. The value obtained, w_s , is then inserted into the pump shaft power equation:

$$\dot{W}_s = \dot{m}w_s$$

to find the total power required by the pump.

66. Net positive suction head, NPSH

$$NPSH_A = \frac{P_{suction}}{\rho \cdot g} + \frac{v_{suction}^2}{2 \cdot g} - \frac{P_v}{\rho \cdot g}$$

For suction from a tank/free surface

$$NPSH_A = \frac{P_{surface}}{\rho \cdot g} + z_{surface} - h_{f,suction} - \frac{P_v}{\rho \cdot g}$$

Cavitation condition:

$$NPSH_A > NPSH_R$$

Where:

- $NPSH_A$ = net positive suction head available (m)
- $NPSH_R$ = net positive suction head required by the pump (m)
- $P_{suction}$ = absolute pressure at pump suction (Pa)
- $v_{suction}$ = fluid velocity at pump suction ($\text{m} \cdot \text{s}^{-1}$)
- P_v = vapour pressure of the liquid at operating temperature (Pa)
- ρ = fluid density ($\text{kg} \cdot \text{m}^{-3}$)
- g = gravitational acceleration ($\text{m} \cdot \text{s}^{-2}$)

- $P_{surface}$ = absolute pressure above the suction tank/free surface (Pa)
- $z_{surface}$ = elevation head of liquid surface relative to pump centreline (m)
- $h_{f,suction}$ = head loss in the suction line (m)

This is used for:

Checking whether the pressure at the pump inlet is high enough to avoid cavitation. Cavitation happens when the liquid pressure becomes too low and the liquid starts to vaporize inside the pump.

When to use this:

Use when the problem involves pumps, cavitation, suction line, vapour pressure, pump inlet pressure, pump placement, pump suction, or whether a pump can operate safely. Trigger words: NPSH, cavitation, suction head, suction pressure, vapour pressure, pump inlet, pump required head, pump available head.

IMPORTANT:

Use absolute pressure, not gauge pressure, in NPSH calculations.

The pump is safe from cavitation when:

$$NPSH_A > NPSH_R$$

Usually, $NPSH_A$ should be larger than $NPSH_R$ with a safety margin.

67. Pump shaft power

$$\dot{W}_s = \dot{m} \cdot w_s$$

Where:

- $\dot{W}_s$ = shaft power (W)
- $\dot{m}$ = mass flow rate (kg s^{-1})
- w_s = specific shaft work (J kg^{-1})

This is used for:

Calculating the mechanical power required by pumps. Typical questions involve determining pump energy requirements during fluid transport.

When to use this:

Use when calculating total mechanical power required by a pump. Trigger words: pump power, energy requirement, shaft power, motor sizing.

68. Pump hydraulic power

$$P_{hyd} = \Delta P \cdot Q$$

Or

$$P_{hyd} = \rho \cdot g \cdot Q \cdot H$$

Where:

- P_{hyd} = hydraulic power transferred to the liquid ($\text{W} = \text{J s}^{-1}$)
- ΔP = pressure increase across pump (Pa)
- Q = volumetric flow rate ($\text{m}^3 \text{s}^{-1}$)
- ρ = fluid density (kg m^{-3})
- g = gravitational acceleration (m s^{-2})
- H = pump head (m)

This is used for:

Calculating the useful power transferred by a pump to the fluid.

When to use this:

Use when flowrate and pressure increase or pump head are given. Trigger words: hydraulic power, pump head, pressure increase, flowrate.

69. Pump efficiency

$$\eta = \frac{P_{hyd}}{P_{shaft}}$$

Where:

- η = pump efficiency, dimensionless
- P_{hyd} = hydraulic power transferred to the liquid (W)
- P_{shaft} = shaft/motor power supplied to the pump (W)

This is used for:

Relating useful hydraulic power to the actual power supplied to the pump.

When to use this:

Use when a problem asks for pump efficiency or gives both hydraulic power and shaft power. Trigger words: efficiency, useful power, input power, output power, pump performance.

IMPORTANT:

If efficiency is given as a percentage, convert it to a decimal before using it: (e.g. 80%=0.80).

70. Pump shaft power with efficiency

$$P_{shaft} = \frac{P_{hyd}}{\eta}$$

Or

$$P_{shaft} = \frac{\rho \cdot g \cdot Q \cdot H}{\eta}$$

Where:

- P_{shaft} = actual shaft/motor power required (W)
- P_{hyd} = hydraulic power transferred to liquid (W)
- η = pump efficiency (*dimensionless*)
- ρ = fluid density (kg m^{-3})
- g = gravitational acceleration (m s^{-2})
- Q = volumetric flow rate ($\text{m}^3 \text{s}^{-1}$)
- H = pump head (m)

This is used for:

Calculating actual pump power when pump efficiency is included.

When to use this:

Use when the problem gives pump efficiency. Trigger words: pump efficiency, actual shaft power, motor power, efficiency.

71. Energy consumption over time

$$E = P \cdot t$$

Where:

- E = energy consumed (J)
- P = power ($\text{W} = \text{J s}^{-1}$)
- t = operating time (s)

This is used for:

Calculating total energy use from a known power demand over time.

When to use this:

Use when asked for total energy consumption, electricity use, or operating cost. Trigger words: energy consumption, kWh, operating time, power used for X hours.

Useful conversion:

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

72. Tank emptying time

$$t = \frac{V}{Q}$$

Where:

- t = time (s)
- V = volume (m^3)
- Q = volumetric flow rate (m^3s^{-1}) (can also have the letter F)

This is used for:

Calculating the time required to fill or empty tanks. Typical questions combine this equation with flow-rate calculations.

When to use this:

Use when calculating filling or emptying duration from known flow rate. Trigger words: emptying time, filling time, how long will the tank take, drainage duration.

73. Reynolds number (Stirred tank)

$$Re = \frac{\rho \cdot N \cdot D_i^2}{\mu}$$

Where:

- Re = Reynolds number (dimensionless)
- ρ = fluid density (kg m^{-3})
- N = impeller rotational speed (s^{-1})
- D_i = impeller diameter (m)
- μ = dynamic viscosity (Pa s)

This is used for:

Determining the mixing regime in stirred bioreactors. Typical questions use this to evaluate whether mixing is turbulent enough for efficient oxygen transfer and suspension mixing.

When to use this:

Use to characterize mixing regime in stirred bioreactors. Trigger words: mixing regime, agitation, impeller flow, stirred tank hydrodynamics.

IMPORTANT:

Reynolds number determination of flow for stirred tank: : laminar, transitional or turbulent.

Flow regime:

Reynolds number	Flow regime
$Re < 10$	Laminar
$10 < Re < 10^4$	Transitional
$Re > 10^4$	Turbulent

74. Tip speed equation

$$u_{tip} = \pi \cdot N \cdot D_i$$

Where:

- u_{tip} = impeller tip speed (m s^{-1})
- N = impeller rotational speed (s^{-1})
- D_i = impeller diameter (m)

This is used for:

Scale-up and mixing calculations in stirred bioreactors. Tip speed is important because it affects shear stress and mixing intensity.

When to use this:

Use when evaluating shear stress and mixing intensity from impeller rotation. Trigger words: tip speed, shear sensitivity, cell damage, impeller speed.

75. Power input and mixing

$$P = N_p \cdot \rho \cdot N^3 \cdot D_i^5$$

Where:

- P = power input (W)
- N_p = impeller power number (dimensionless)
- ρ = fluid density (kg m^{-3})
- N = impeller rotational speed (s^{-1})
- D_i = impeller diameter (m)

It is used when calculating mixing intensity and scale-up effects.

This is used for:

Increasing agitation increases turbulence and bubble breakup, which usually increases oxygen transfer.

When to use this:

Use for mixing, agitation, and oxygen transfer scale-up calculations. Trigger words: power consumption, mixing intensity, agitation power, scale-up, $k_L a$ correlation.

This is often connected to $k_L a$ - Power input is often related empirically to oxygen transfer:

$$k_L a \propto \left(\frac{P}{V}\right)^a \cdot v_s^b$$

Where:

- $k_L a$ = volumetric oxygen transfer coefficient (h^{-1})
- P/V = power input per reactor volume (W m^{-3})
- v_s = superficial gas velocity (m s^{-1})
- a, b = empirical constants (dimensionless)

Interpretation:

- Higher stirring speed $\rightarrow$ larger $P \rightarrow$ higher $k_L a$
- Higher aeration $\rightarrow$ larger $v_s \rightarrow$ higher $k_L a$
- Better oxygen transfer supports higher biomass concentrations and productivity.

This is used for:

This relationship is important because increasing agitation and aeration generally increases oxygen transfer, allowing higher biomass concentrations and productivity in aerobic bioprocesses.

76. Volumes of gas per volume liquid per minute, vvm

$$vvm = \frac{Q_g}{V_L}$$

Where:

- vvm = aeration rate, gas volume per liquid volume per minute (min^{-1})
- Q_g = gas volumetric flow rate ($\text{m}^3 \text{min}^{-1}$)
- V_L = liquid volume (m^3)

This is used for:

Expressing aeration rate in fermenters.

When to use this:

Use when aeration is given relative to liquid volume. Trigger words: vvm, air flowrate, aeration rate, sparging.

77. Gas flowrate from vvm (aeration rate)

$$Q_g = vvm \cdot V_L$$

Where:

- Q_g = gas volumetric flow rate ($\text{m}^3 \text{min}^{-1}$)
- vvm = aeration rate (min^{-1})
- V_L = liquid volume (m^3)

This is used for:

Calculating required air/gas flowrate from a specified aeration rate.

When to use this:

Use when the problem gives vvm and asks for air/gas flowrate. Trigger words: calculate air flow, vvm, sparging rate, gas flowrate.

78. Aeration number / gas flow number, N_A

$$N_A = \frac{Q_g}{N \cdot D_i^3}$$

Where:

- N_A = aeration number / gas flow number, dimensionless
- Q_g = gas volumetric flow rate ($\text{m}^3 \text{s}^{-1}$)
- N = stirrer speed (s^{-1})
- D_i = impeller diameter (m)

This is used for:

Describing the gas flow rate relative to impeller speed and impeller size in an aerated stirred tank. It is used to assess gas dispersion and risk of impeller flooding.

When to use this:

Use when a problem involves aerated stirred tanks, gas dispersion, or impeller flooding, especially when gas flow rate, impeller diameter, and stirrer speed are involved. Trigger words: prevent impeller flooding, Rushton turbine, gas dispersion, air flow rate, sparging, aerated stirred tank, flooding criterion.

IMPORTANT NOTE

For Rushton turbines, the rule

$$N_A \approx 0.1$$

As a flooding limit is typically used as a simplified criterion.

BUT

If the question provides a specific flooding-loading correlation, that correlation should be used if it is intended. E.g. from exam set spring 2025 where “rushton turbine” is used BUT

$$Fl_g = 30 \left(\frac{D_i}{D_T} \right)^{3.5} Fr$$

and

$$Fl_g = 0.2 \left(\frac{D_i}{D_T} \right)^{0.5} Fr^{0.5}$$

Was given – then using the above simplification may be too simplified.

79. Gas flow number for flooding correlations, Fl_g

$$Fl_g = \frac{Q_g}{N_A \cdot D_i^3}$$

Where:

- Fl_g = gas flow number, dimensionless (also written as N_A , see point 61)
- Q_g = gas volumetric flow rate ($\text{m}^3 \text{s}^{-1}$)
- N = stirrer speed (s^{-1})
- D_i = impeller diameter (m)

This is used for:

Describing the gas flow rate relative to impeller speed and impeller size in an aerated stirred tank.

When to use this:

Use when the problem uses the notation Fl_g , or when gas loading, flooding, loading, or complete gas dispersion is described using gas flow number. In most cases in this course, Fl_g is equivalent to the aeration number N_A , because both are defined as $Q_g/(ND_i^3)$. Always follow the notation and definition given in the question. Trigger words: gas flow number, flooding-loading transition, complete gas dispersion, aerated stirred tank, Rushton turbine.

80. Stirrer speed required to prevent flooding

$$N = \frac{Q_g}{N_A \cdot D_i^3}$$

Where:

- N = required stirrer speed (s^{-1})
- Q_g = gas volumetric flow rate ($\text{m}^3 \text{s}^{-1}$)
- N_A = critical aeration number for flooding, dimensionless
- D_i = impeller diameter (m)

This is used for:

Calculating the minimum stirrer speed needed to prevent impeller flooding in an aerated stirred tank.

When to use this:

Use when the problem asks whether an impeller will flood or asks for the stirrer speed required to prevent flooding. For a Rushton turbine, a common approximate flooding criterion is $N_A \approx 0.1$, unless another value is given. Trigger words: minimum stirrer speed, prevent flooding, impeller flooding, gas flow number, critical aeration number, Rushton turbine.

81. Stirrer speed unit conversion

$$N_{rpm} = 60 \cdot N$$

Where:

- N_{rpm} = stirrer speed (rpm)
- N = stirrer speed (s^{-1})

This is used for:

Converting stirrer speed from revolutions per second to revolutions per minute.

When to use this:

Use when the calculated stirrer speed is in s^{-1} but the final answer is requested in rpm. Trigger words: rpm, revolutions per minute, stirrer speed, rotational speed.

82. Froude number for stirred tanks

$$Fr = \frac{N^2 \cdot D_i}{g}$$

Where:

- Fr = Froude number, dimensionless
- N = stirrer speed (s^{-1})
- D_i = impeller diameter (m)
- g = gravitational acceleration (m s^{-2})

This is used for:

Comparing inertial forces from impeller rotation with gravitational forces in stirred tanks.

When to use this:

Use when gas dispersion, flooding, loading, vortex formation, or stirred-tank scale-up correlations include Fr . Trigger words: Froude number, flooding-loading transition, complete gas dispersion, gas dispersion correlation, surface vortex, aerated stirred tank, stirred tank.

83. Froude number for non-stirred/free-surface flow

$$Fr = \frac{v}{\sqrt{g \cdot L}}$$

Or equivalently

$$Fr^2 = \frac{v^2}{g \cdot L}$$

Where:

- Fr = Froude number, dimensionless
- v = characteristic fluid velocity (m s^{-1})
- g = gravitational acceleration (m s^{-2})
- L = characteristic length, e.g. hydraulic depth, tank length, pipe diameter, or liquid depth depending on the system (m)

This is used for:

Comparing inertial forces with gravitational forces in non-stirred systems, especially free-surface or gravity-dominated flow.

When to use this:

Use only if the problem concerns non-stirred free-surface flow, open-channel flow, waves, hydraulic jumps, or gravity-dominated flow behaviour. It is usually not needed for standard stirred bioreactor calculations. Trigger words: free-surface flow, open-channel flow, waves, hydraulic jump, gravity-dominated flow, flow over liquid surface.

84. Impeller types and flow pattern**Where:**

- Rushton turbine = radial-flow impeller, commonly used for gas dispersion in aerobic fermentation
- Pitched-blade turbine = mixed axial/radial-flow impeller, commonly used for general mixing and suspension
- Marine propeller = axial-flow impeller, commonly used for low-viscosity liquid circulation

- Hydrofoil impeller = axial-flow impeller, commonly used for efficient large-scale mixing with lower shear
- Anchor/helical ribbon impeller = tangential/laminar-flow impeller, commonly used for high-viscosity broths

This is used for:

Identifying the mixing pattern and expected gas-handling behaviour of the impeller used in a bioreactor.

When to use this:

Use when the problem specifies the impeller type or asks about mixing, aeration, gas dispersion, shear sensitivity, or flooding behaviour. For Rushton turbines, flooding/gas-dispersion problems commonly use the aeration number $N_A = \frac{Q_g}{ND_t^3}$. Trigger words: Rushton turbine, gas dispersion, radial flow, pitched blade, hydrofoil, high-viscosity broth, shear-sensitive cells.

IMPORTANT:

The Rushton turbine is usually the only impeller type where you are expected to know a specific standard assumption without it being given, especially for gas dispersion/flooding problems.

The key assumption is usually:

$$N_A \approx 0.1$$

as an approximate flooding criterion.

For the other impeller types, you usually only need to know the qualitative behaviour, unless the problem gives a specific correlation or value.

85. General reaction rate equation

For reactant/substrate consumption

$$-r_A = -\frac{d \cdot C_A}{dt}$$

Or for product formation

$$r_P = \frac{d \cdot C_P}{dt}$$

Where:

- r_A = reaction rate for reactant A (mol L⁻¹ s⁻¹) or (mol L⁻¹ h⁻¹)
- r_P = product formation rate (mol L⁻¹ s⁻¹) or (mol L⁻¹ h⁻¹)
- C_A = concentration of reactant A (mol L⁻¹)
- C_P = concentration of product P (mol L⁻¹)
- t = time (s) or (h)

This is used for:

Describing how fast a reactant is consumed or a product is formed in a reaction. In bioprocess technology, this is used for enzyme reactions, substrate consumption, product formation, and dynamic reactor balances.

When to use this:

Use when the question asks for reaction rate from concentration change over time. Trigger words: reaction rate, rate of disappearance, rate of formation, concentration-time data, substrate consumption over time, product formation over time

86. General rate law

$$-r_A = k \cdot C_a^n$$

Where:

- $-r_A$ = rate of consumption of reactant A ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- k = reaction rate constant, units depend on reaction order
- C_A = concentration of reactant A (mol L^{-1})
- n = reaction order (dimensionless)

This is used for:

Describing how the reaction rate depends on reactant concentration. The reaction order tells whether the rate is independent of concentration, proportional to concentration, or proportional to concentration squared.

When to use this:

Use when the problem asks about reaction order, rate law, kinetic model, or how concentration affects reaction rate. Trigger words: reaction order, rate constant, rate law, concentration dependence, zero-order, first-order, second-order.

87. Zero-order reaction kinetics

$$-r_A = k$$

Integrated from:

$$C_A = C_{A0} - k \cdot t$$

Linear plot:

$$C_A \text{ vs } t$$

Slope:

$$\text{Slope} = -k$$

Where:

- $-r_A$ = rate of consumption of reactant A ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- k = zero-order rate constant ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- C_A = concentration of reactant A at time t (mol L^{-1})
- C_{A0} = initial concentration of reactant A (mol L^{-1})
- t = time (s) or (h)

This is used for:

Describing reactions where the rate is constant and independent of substrate/reactant concentration. In bioprocesses, this can occur in enzyme-catalysed reactions when the enzyme is saturated with substrate.

When to use this:

Use when the rate stays constant even though concentration changes. Trigger words: zero-order, constant reaction rate, enzyme saturation, substrate in excess, C_A vs time is linear.

88. First-order reaction kinetics

$$-r_A = k \cdot C_A$$

Integrated from:

$$\ln(C_A) = \ln(C_{A0}) - k \cdot t$$

Or:

$$C_A = C_{A0} \cdot e^{-k \cdot t}$$

Linear plot:

$$\ln(C_{A0}) \text{ vs } t$$

Slope:

$$\text{Slope} = -k$$

Where:

- $-r_A$ = rate of consumption of reactant A ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- k = first-order rate constant (s^{-1}) or (h^{-1})
- C_A = concentration of reactant A at time t (mol L^{-1})
- C_{A0} = initial concentration of reactant A (mol L^{-1})
- t = time (s) or (h)

This is used for:

Describing reactions where the rate is proportional to the concentration of one reactant. In bioprocesses, first-order behaviour can occur at low substrate concentration or when substrate becomes limiting.

When to use this:

Use when the reaction slows down as substrate concentration decreases. Trigger words: first-order, exponential decay, concentration-dependent rate, $\ln(C_A)$ vs time is linear, low substrate concentration.

89. Second-order reaction kinetics

$$-r_A = k \cdot C_A^2$$

Integrated from:

$$\frac{1}{C_A} = \frac{1}{C_{A0}} + k \cdot t$$

Linear plot:

$$\frac{1}{C_A} \text{ vs } t$$

Slope:

$$\text{Slope} = k$$

Where:

- $-r_A$ = rate of consumption of reactant A ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- k = second-order rate constant ($\text{L mol}^{-1} \text{s}^{-1}$) or ($\text{L mol}^{-1} \text{h}^{-1}$)
- C_A = concentration of reactant A at time t (mol L^{-1})
- C_{A0} = initial concentration of reactant A (mol L^{-1})
- t = time (s) or (h)

This is used for:

Describing reactions where the rate depends on the square of one reactant concentration or on two reacting species. In bioprocesses, this is less common for simple growth models but can appear in chemical/biochemical reaction kinetics

When to use this:

Use when the reaction rate changes strongly with concentration. Trigger words: second-order, bimolecular reaction, rate proportional to concentration squared, $1/C_A$ vs time is linear.

90. Determining reaction order from experimental data

Zero-order test:

$$C_A \text{ vs } t \rightarrow \text{straight line}$$

First-order test:

$$\ln(C_A) \text{ vs } t \rightarrow \text{straight line}$$

Second-order test:

$$\frac{1}{C_A} \text{ vs } t \rightarrow \text{straight line}$$

Where:

- C_A = concentration of reactant A (mol L⁻¹)
- t = time (s) or (h)

This is used for:

Identifying the reaction order from concentration-time data. The plot that gives the best straight line indicates the reaction order.

When to use this:

Use when given experimental concentration-time data and asked to find the reaction order or rate constant. Trigger words: determine reaction order, kinetic data, concentration versus time, linear plot, graphical method, best straight line.

91. Average reaction rate from concentration-time data

$$-r_A \approx \frac{\Delta C_A}{\Delta t}$$

Where:

- $-r_A$ = average rate of consumption of reactant A (mol L⁻¹ s⁻¹) or (mol L⁻¹ h⁻¹)
- ΔC_A = change in concentration of reactant A (mol L⁻¹)
- Δt = time interval (s) or (h)

This is used for:

Estimating reaction rate between two measured concentration points. It is useful when the rate is not directly given but concentration measurements over time are available.

When to use this:

Use when concentration is measured at different times and the question asks for average rate. Trigger words: average rate, concentration-time table, experimental data, rate between two points, $\Delta C/\Delta t$.

92. Michaelis–Menten equation

$$v = \frac{v_{\max} \cdot S}{K_M + S}$$

Where:

- v = enzyme reaction rate (mol L⁻¹ s⁻¹) or (mol L⁻¹ h⁻¹)
- $v_{\max}$ = maximum enzyme reaction rate (mol L⁻¹ s⁻¹) or (mol L⁻¹ h⁻¹)
- S = substrate concentration (mol L⁻¹)
- K_M = Michaelis constant (mol L⁻¹)

This is used for:

Describing enzyme-catalysed reaction rates as a function of substrate concentration. The rate increases with substrate concentration but approaches $v_{\max}$ when the enzyme becomes saturated.

When to use this:

Use for enzyme kinetics and biocatalysis problems where substrate concentration affects reaction rate. Trigger words: Michaelis–Menten, enzyme kinetics, $v_{\max}$, K_M , enzyme saturation, substrate concentration, biocatalysis.

93. Michaelis–Menten low-substrate approximation

When:

$$S \ll K_M$$

Then:

$$v \approx \frac{v_{max}}{K_M} \cdot S$$

Where:

- v = enzyme reaction rate ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- v_{max} = maximum enzyme reaction rate ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- K_M = Michaelis constant (mol L^{-1})
- S = substrate concentration (mol L^{-1})

This is used for:

Simplifying Michaelis–Menten kinetics at low substrate concentration. Under these conditions, the reaction behaves approximately as first-order with respect to substrate.

When to use this:

Use when substrate concentration is much smaller than K_M . Trigger words: low substrate concentration, $S \ll K_M$, first-order enzyme kinetics, substrate-limited enzyme reaction.

94. Michaelis–Menten high-substrate approximation

When:

$$S \gg K_M$$

Then:

$$v \approx v_{max}$$

Where:

- v = enzyme reaction rate ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- v_{max} = maximum enzyme reaction rate ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- S = substrate concentration (mol L^{-1})
- K_M = Michaelis constant (mol L^{-1})

This is used for:

Simplifying Michaelis–Menten kinetics at high substrate concentration. Under these conditions, the enzyme is saturated and the reaction behaves approximately as zero-order with respect to substrate.

When to use this:

Use when substrate concentration is much larger than K_M . Trigger words: high substrate concentration, $S \gg K_M$, enzyme saturation, zero-order enzyme kinetics, maximum rate.

95. Lineweaver–Burk equation

$$\frac{1}{v} = \frac{K_M}{v_{max}} \cdot \frac{1}{S} + \frac{1}{v_{max}}$$

Linear plot:

$$\frac{1}{v} \text{ vs } \frac{1}{S}$$

Slope:

$$\text{Slope} = \frac{K_M}{v_{max}}$$

Y-intercept:

$$y - \text{intercept} = \frac{1}{v_{max}}$$

X-intercept:

$$x - \text{intercept} = -\frac{1}{K_M}$$

Where:

- v = enzyme reaction rate ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- $v_{\max}$ = maximum enzyme reaction rate ($\text{mol L}^{-1} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)
- S = substrate concentration (mol L^{-1})
- K_M = Michaelis constant (mol L^{-1})

This is used for:

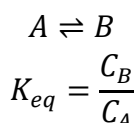
Linearising Michaelis–Menten data to estimate $v_{\max}$ and K_M from enzyme kinetic experiments.

When to use this:

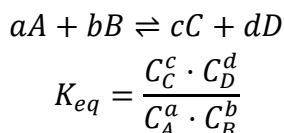
Use when given enzyme kinetic data and asked to determine $v_{\max}$ and K_M from a double-reciprocal plot. Trigger words: Lineweaver–Burk, double reciprocal plot, enzyme kinetics graph, estimate K_M , estimate $v_{\max}$, $1/v$ vs $1/S$.

96. Equilibrium constant for reversible reaction

For:



For a general reaction:



Where:

- = equilibrium constant (dimensionless) if concentrations are made dimensionless by a standard state
- C_A, C_B, C_C, C_D = concentrations of species A, B, C, D (mol L^{-1})
- a, b, c, d = stoichiometric coefficients (dimensionless)

This is used for:

Describing the final equilibrium composition of reversible reactions. In biocatalysis, this is used when a reaction does not go fully to completion and both reactant and product are present at equilibrium.

When to use this:

Use when the question mentions equilibrium, reversible reaction, final composition, conversion at equilibrium, or equilibrium constant. Trigger words: equilibrium constant, reversible reaction, final concentration, reaction equilibrium, K_{eq} .

97. Dynamic component mass balance in a reactor

$$V \cdot \frac{dC_i}{dt} = F_{in} \cdot C_{i,in} - F_{out} \cdot C_i + r_i \cdot V$$

For constant volume and equal inlet/outlet flow:

$$\frac{dC_i}{dt} = D(C_{i,in} - C_i) + r_i$$

Where:

- V = reactor volume (L) or (m^3)
- C_i = concentration of component i in reactor/outlet (mol L^{-1}) or (g L^{-1})
- $C_{i,in}$ = inlet concentration of component i (mol L^{-1}) or (g L^{-1})
- F_{in} = inlet volumetric flow rate (L h^{-1}) or ($\text{m}^3 \text{s}^{-1}$)
- F_{out} = outlet volumetric flow rate (L h^{-1}) or ($\text{m}^3 \text{s}^{-1}$)

- r_i = reaction rate of component i , positive for formation and negative for consumption (mol L⁻¹ h⁻¹) or (g L⁻¹ h⁻¹)
- D = dilution rate (h⁻¹)

This is used for:

Writing time-dependent mass balances for reacting components in reactors. It combines flow in, flow out, and reaction formation/consumption.

When to use this:

Use when concentration changes with time in a reactor and reaction is included. Trigger words: dynamic mass balance, unsteady state, accumulation, reacting system, concentration over time, CSTR with reaction.

98. Batch reactor balance with reaction

$$\frac{dC_i}{dt} = r_i$$

For substrate consumption:

$$\frac{dS}{dt} = r_s$$

Where $r_s < 0$.

For product formation:

$$\frac{dP}{dt} = r_p$$

Where $r_p > 0$.

Where:

- C_i = concentration of component i (mol L⁻¹) or (g L⁻¹)
- S = substrate concentration (mol L⁻¹) or (g L⁻¹)
- P = product concentration (mol L⁻¹) or (g L⁻¹)
- r_i = reaction rate of component i (mol L⁻¹ h⁻¹) or (g L⁻¹ h⁻¹)
- r_s = substrate consumption rate (mol L⁻¹ h⁻¹) or (g L⁻¹ h⁻¹)
- r_p = product formation rate (mol L⁻¹ h⁻¹) or (g L⁻¹ h⁻¹)
- t = time (h)

This is used for:

Describing concentration changes in a batch reactor where there is no inlet or outlet flow. Changes are caused only by reaction.

When to use this:

Use for batch reactions where concentration changes over time and there is no flow in or out. Trigger words: batch reactor, no inlet, no outlet, concentration-time profile, reaction only, dynamic batch balance.

99. Dynamic CSTR component balance

$$V \cdot \frac{dC_A}{dt} = FC_{A,in} - FC_A + r_A \cdot V$$

Where:

- V = reactor volume (m³) or (L)
- C_A = concentration of component A in reactor/outlet (mol m⁻³) or (mol L⁻¹)
- $C_{A,in}$ = inlet concentration of component A (mol m⁻³) or (mol L⁻¹)
- t = time (s) or (h)
- F = volumetric flow rate (m³ s⁻¹) or (L h⁻¹)

- r_A = reaction rate of component A ($\text{mol m}^{-3} \text{s}^{-1}$) or ($\text{mol L}^{-1} \text{h}^{-1}$)

This is used for:

Describing how the concentration of a component changes over time in a continuously stirred tank reactor when there is inlet flow, outlet flow, and reaction.

When to use this:

Use for time-dependent concentration changes in a CSTR or well-mixed continuous bioreactor. Trigger words: dynamic CSTR, unsteady-state reactor, concentration changes with time, inlet concentration, outlet concentration, continuous stirred tank, reactor balance.

100. Filtration rate equation

$$\frac{dV}{dt} = \frac{\Delta P \cdot A}{\mu \cdot (R_m + R_c)}$$

Where:

- $\frac{dV}{dt}$ = filtration flow rate ($\text{m}^3 \text{s}^{-1}$)
- V = filtrate volume (m^3)
- t = filtration time (s)
- ΔP = pressure difference across filter (Pa)
- A = filter area (m^2)
- μ = dynamic viscosity of filtrate (Pa s)
- R_m = filter medium resistance (m^{-1})
- R_c = cake resistance (m^{-1})

This is used for:

Calculating the rate of filtration through a filter medium and filter cake under pressure-driven filtration.

When to use this:

Use when the problem asks for filtration rate, filtrate flow, required filter area, or pressure-driven separation of solids from liquid. Trigger words: filtration, filter area, pressure drop across filter, filtrate volume, filter cake, medium resistance, cake resistance.

NOTE:

Cake resistance is the flow resistance caused by the accumulated solid cake on the filter surface. It increases as more solids are deposited during filtration, because the liquid must pass through a thicker and more compact particle layer. High cake resistance gives a lower filtration rate and may require a larger filter area, higher pressure difference, or longer filtration time.

101. Cake resistance in filtration

$$R_c = \alpha \cdot \frac{C_s \cdot V}{A}$$

Where:

- R_c = cake resistance (m^{-1})
- α = specific cake resistance (m kg^{-1})
- C_s = mass of dry solids deposited per filtrate volume (kg m^{-3})
- V = filtrate volume (m^3)
- A = filter area (m^2)

This is used for:

Calculating how the resistance of the filter cake increases as more solids accumulate during filtration.

When to use this:

Use when cake filtration is involved and the filter cake resistance changes with filtrate volume. Trigger words: cake filtration, specific cake resistance, solids concentration, accumulated cake, increasing filtration resistance.

NOTE:

A filter cake is the layer of retained solid particles that builds up on the filter surface during filtration. In bioprocessing, this can be biomass cells, cell debris, precipitated proteins, or other suspended solids from the broth. As the cake becomes thicker, it increases the resistance to liquid flow and lowers the filtration rate.